Experiment No: 03

Name of the Experiment: Clipper and Clamper circuits.

Objective:

Study of Clipper and Clamper circuits.

Theory:

Clipper: Clippers remove signal voltage above and below a specified level. In the experiment no. 2, half wave rectifier can also be called as a clipper circuit. Because it clipped off the negative half cycle of the input signal.

A diode connected in series with the load can clipped off any half cycle of input depending on the orientation of the diode. (Figure 4.1) -

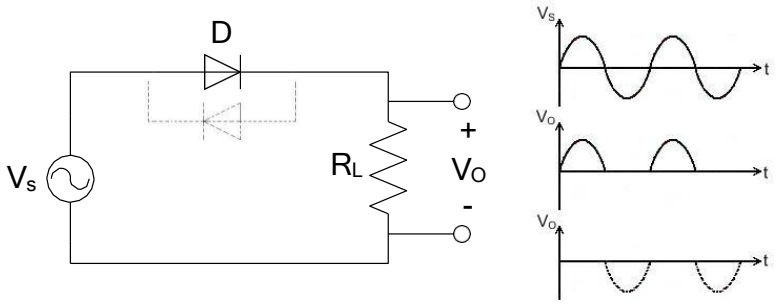


Figure 4.1: Simple Diode Clipper circuit

It is also possible to clip off a certain part of the input signal bellow a specified signal level by using a voltage source in reverse bias condition with the diode. If a battery of V volts is added to it, then for V_s above (V+0.7) volts the diode becomes forward bias and turns ON. The load receives above this voltage Level.

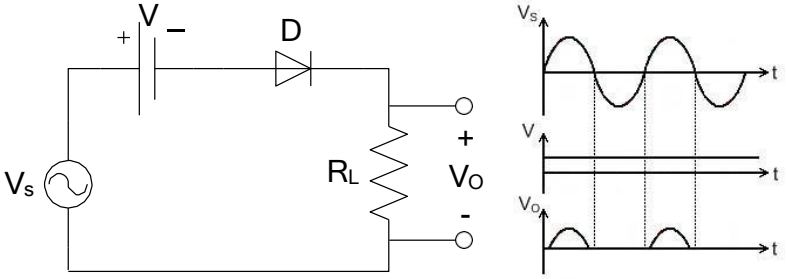


Figure 4.2: Clipper Circuit Using Bias Diode.

A diode connected in parallel with the load can clip off the input signal above 0.7 volts of one-half cycle depending on the connection of the diode. Using two diodes in parallel in opposite direction both the half cycle can be limited to 0.7 volts.

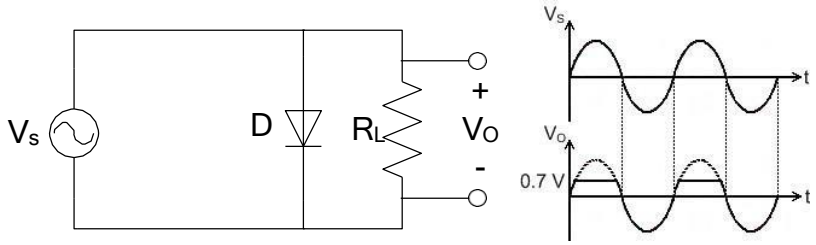


Figure 4.3: Parallel Clipper Circuit.

Using a biased diode it is possible to limit the output voltage to a specified level depending on the attached battery voltage. Either the half cycles or both of them can be clipped off above a specified level.

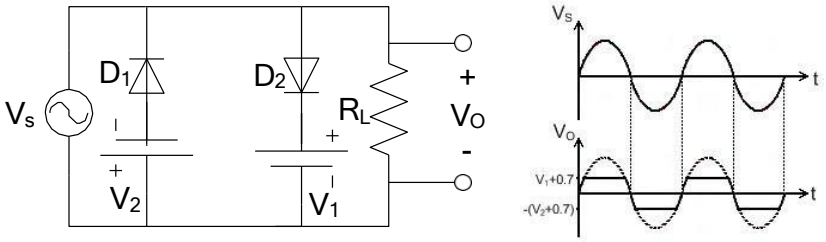


Figure 4.4: Biased Parallel Clipper Circuit.

In practical case for both the series and parallel clippers voltage source is not added. Required voltage levels are maintained by adding more semiconductor diode.

Clamper: A DC clamper circuit adds a DC voltage to the input signal. For instance, if the incoming signal varies from -10 volts to +10 volts, a positive DC clamper will produce an output that ideally swing from 0 volts to 20 volts and a negative clamper would produce an output between 0 volts to -20 volts.

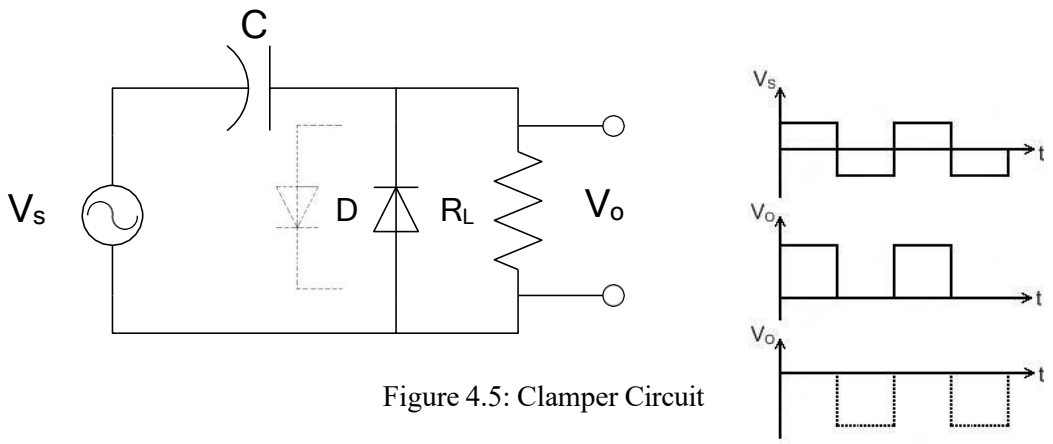


Figure 4.5: Clamper Circuit

Equipment and Components:

Serial no.	Component Details	Specification	Quantity
1.	p-n junction diode	1N4007	1 piece
2.	Resistor	100KΩ	1 piece
3.	Capacitor	0.1μF	1 piece
4.	Signal generator		1 unit
5.	Trainer Board		1 unit
6.	DC power Supply		1 unit
7.	Oscilloscope		1 unit
8.	Digital Multimeter		1 unit
9.	Chords and wire		as required

Experimental Setup:

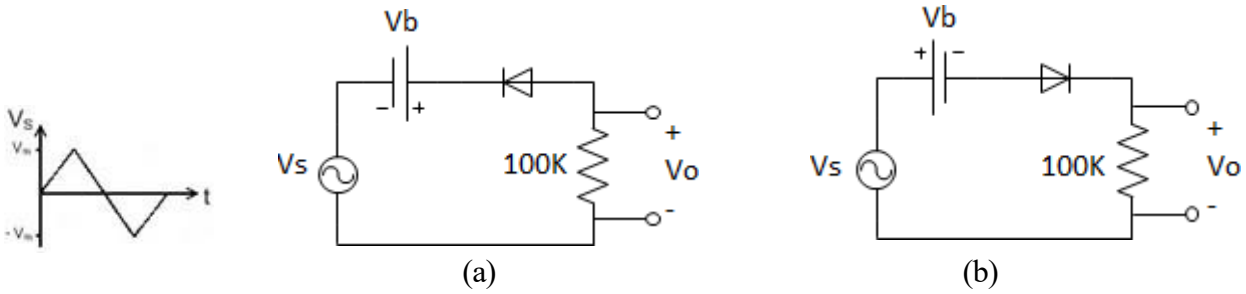


Figure 4.6: Series Clipper Circuit

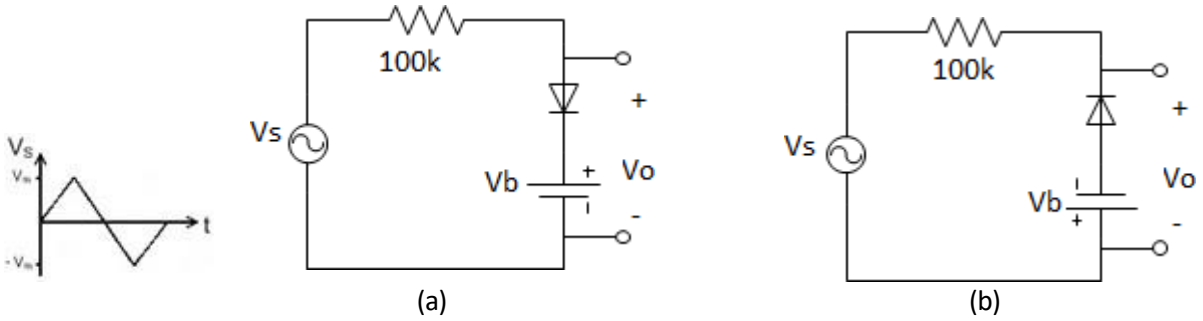


Figure 4.7: Parallel Clipper Circuits

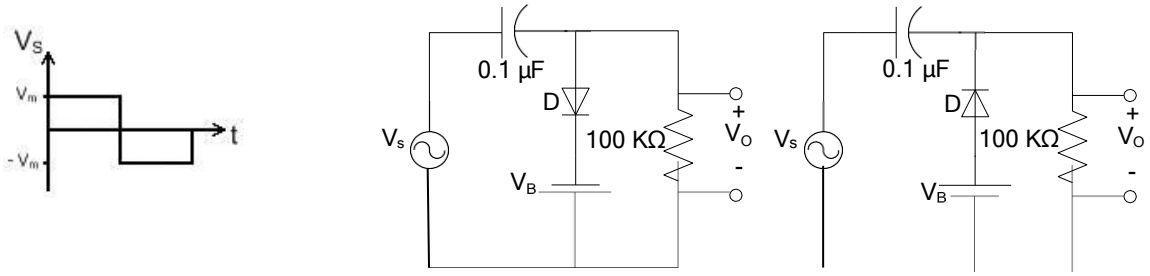


Figure 4.8: Clamper circuit

Procedure:

1. Connect the circuit as shown in the figure 4.6.
2. Using Signal generator, apply a 1kHz $10V_{p\text{-}p}$ sinusoidal voltage source input ($V_m = 5V$)
3. Fix V_b to 2.5V and In the same graph paper, Draw V_s and V_o.
4. Decrease the value of V_b from 2.5V to 0V, and observe the output wave shapes
5. Increase the value of V_b from 2.5V to 5V, and observe the output wave shapes
6. Repeat step 2-4 for figure 3.7 and figure 3.8
7. Record V_{max} and V_{min} for the output wave for the clamper circuit only for V_b=2.5v.

Data Collection:

Signature of instructor:

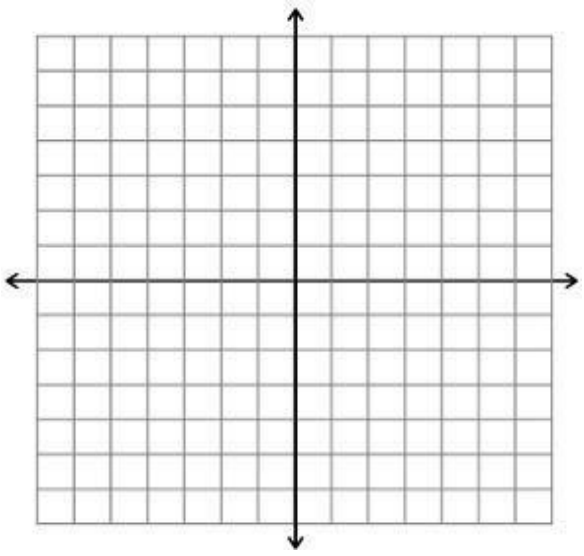
Experiment: 4,
Performed by Group# _____

Theoretical value: R = ____ kΩ

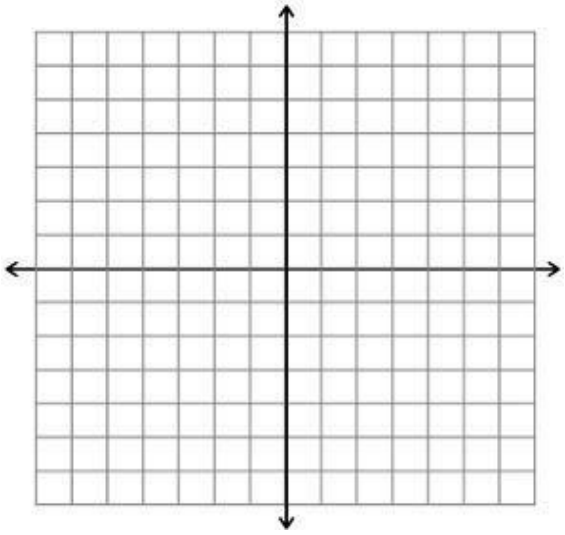
Measured value: R = ____ kΩ

Vs = _____V(p-p).

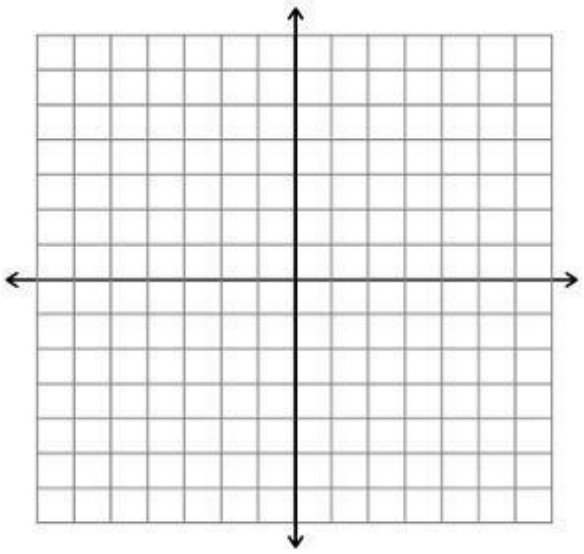
V_b (V)	Vo (p-p)					
	Fig 3.6		Fig 3.7		Fig 3.8	
	(a)	(b)	(a)	(b)	(a)	(b)
0						
1						
2						
2.5					V_{max}= V_{min}=	V_{max}= V_{min}=
3						
4						
5						



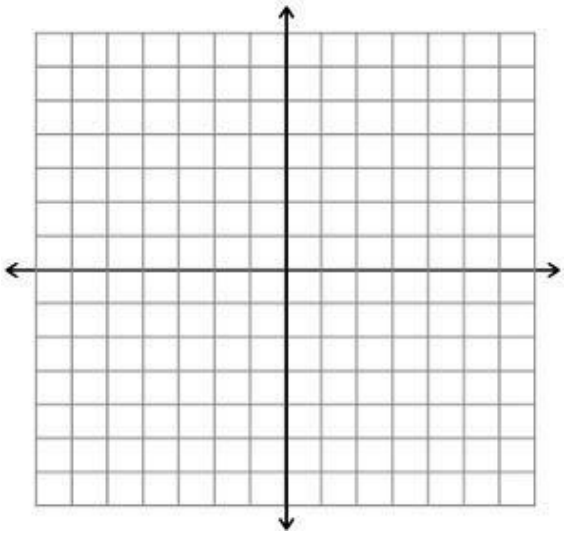
Input-output Fig 4.6 (a)



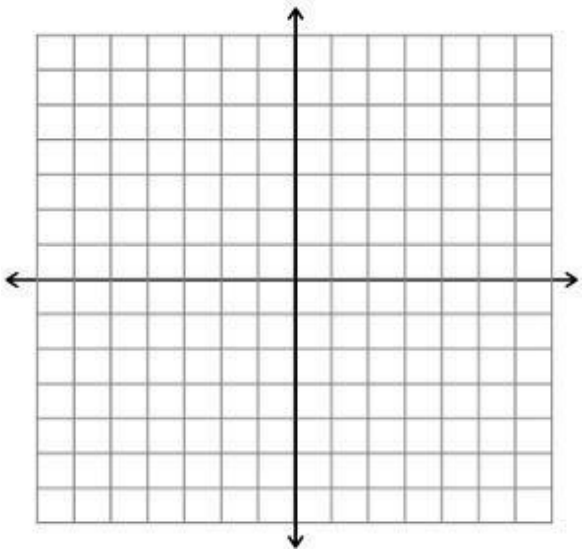
Input-output Fig 4.6 (b)



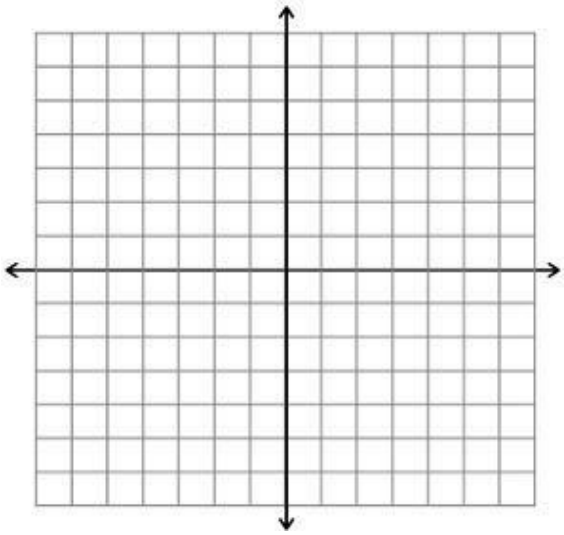
Input-output Fig 4.7 (a)



Input-output Fig 4.7 (b)



Input-output Fig 4.8 (a)



Input-output Fig 4.8 (b)

Report:

1. Using values from your data table, for all the circuit diagrams plot the input-output waveforms observed on the oscilloscope for V_b=2.5V.
2. For Fig 4.6(a &b), Fig4.7 (a & b) and Fig 4.8 (a & b) what change did you observe in the output voltage, In procedure-4? *Explain the reason behind such a change.*
3. For Fig 4.6(a &b), Fig4.7 (a & b) and Fig 4.8 (a & b) what change did you observe in the output voltage, In procedure-5? *Explain the reason behind such a change.*
4. Compare series and parallel clipper circuits.